New NASA spacesuits
In April, NASA gave the public a chance to vote on its next spacesuit from among 3 designs **http://dgit.in/1iqvPcG**

Exoplanet similar to earth
Kepler-186f turns out to be perfect for life being in the goldilocks zone around star Kepler 186 **http://dgit.in/Kepler186-F**

Interstellar Imagination: Spacecraft designed with Infinity in Mind

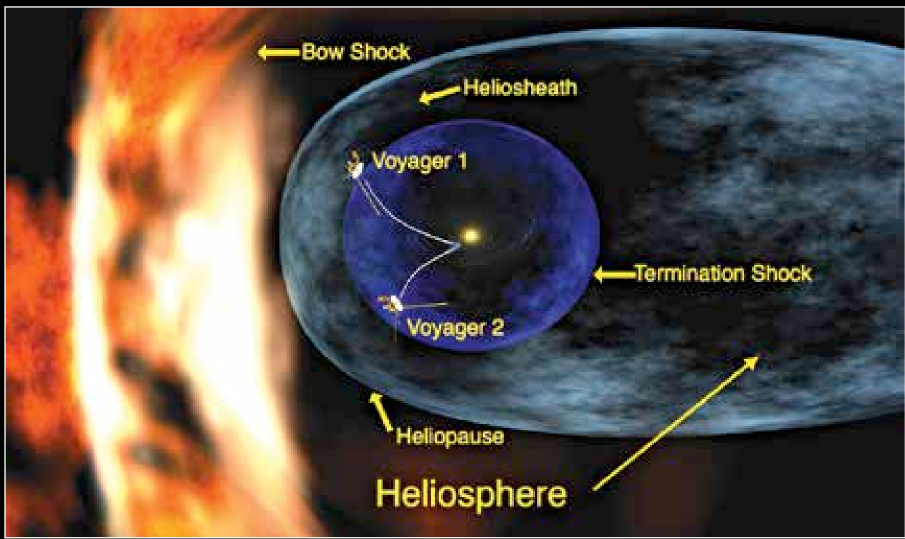
What kind of exotic ships will let us truly reach for the stars?

Samir Alam
readersletters@thinkdigit.com

"Ship like this, be with you till the day you die."
- Captain Mal, Firefly
(Tim Minear/Joss Whedon)

On 24 August 2013, the Voyager I space probe became the first man made object to journey beyond the heliopause - the ecosystem boundary of our Sun. The epic journey took 35 years 11 months and 21 days, and finally introduced humanity to the interstellar Universe - a part of the cosmos beyond all human experience, only ever known through telescopes and radio spectroscopy. We had finally crossed the threshold into the undiscovered country.

And if we as a species are capable of going from Robert Goddard's 1912 test flight of the first liquid-fuel propelled rockets to discovering an Earth like planet such as Kepler 186f in 2013, then the next 100 years can only take us farther. Soon it will be time once again to venture forth boldly, with no data to support our ambition nor clear view of where we are heading, the only undeniable truth will be that we should begin - now.



Over 11000 human work-years were dedicated to just get Voyager to Neptune.

Designing Perfection - From the Imagination to the Cosmos

Long before the idea of the airplane or journey's beyond the planet could be envisioned by science, we had writers and artists prophesying its possibility. It was the musings of Jules Verne in his 1865 novel "From the Earth to the Moon" that gave Konstantin Tsiolkovsky - the founding father of rocketry and astronautics - the inspiration to develop rocket propulsion. And in that spirit, the work being done by groups such as the 100 Year Starship programme holds true promise. The program was initiated by the United States Defense Advanced Research Projects Agency (DARPA) and the National Aeronautics and Space Administration (NASA) in 2012 with a simple purpose - to foster the quest to achieve interstellar space travel in the next 100 years.

This quest is fuelled by the vision already realised in our collective imagination through stories. In a case of reality imitating art, the imaginary inventions of science fiction writers have given us a peak into the future. The designs and technologies, imagined in the great stories of our contemporary history are a treasure trove of scientific possibilities playing themselves out. In some cases the realities constructed are pure fantasy but in others they come very close to science.

One such contemporary example of a scientifically accurate starship is the fictional ISV Venture Star from James Cameron's movie Avatar. The interstellar spacecraft follows all the factual standards of science needed for deep space travel with none of the fictional stylisms like sleek aerodynamics or faster than light travel. The idea of the Venture Star was based on the designs of polymath Charles Pellegrino and physicist Jim Powell who conceived it originally as a craft called the Valkyrie Antimatter Starship in their 1993 novel "Flying to Valhalla". The ISV Venture Star in Avatar may represent the absurdly evil Earth based corporations of the future but its technological foundation is rock solid in today's science.

This scientific authenticity of the Venture Star's design lies in the fact that its propulsion engines are placed on the nose